ENGINEERING PHYSICS | B.TECH 1ST YEAR | SEMESTER I

Wave Optics

Complete Formula & Concept Sheet

Unit I – Interference | Diffraction | Polarization

Includes: All Formulas | Comparison Tables | Flashcards | Exam Tips

Quick Revision Sheet | Symbol Legend

Last-day revision weapon — scan, memorize, score.

1. Course Objectives

- Understand the wave nature of light and its behavior in different media.
- Analyze interference, diffraction, and polarization phenomena quantitatively.
- Apply optical principles to real systems (Newton's rings, gratings, Nicol's prism).

2. Interference – Fundamentals

2.1 Principle of Superposition

When two or more waves overlap at a point, the resultant displacement equals the algebraic sum of individual displacements.

$$y = y_1 + y_2$$

2.2 Conditions for Sustained Interference

Necessary Conditions

- Sources must be coherent (constant phase difference).
- Same frequency (monochromatic preferred).
- Nearly equal amplitudes for good contrast.
- Waves must be in the same state of polarization.

Resultant Intensity

$$I = I_1 + I_2 + 2\sqrt{(I_1 I_2)}\ \cos\delta$$

where δ = phase difference between the two waves

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

2.3 Path Difference and Phase Difference

$$\delta\ (\text{phase}) = (2\pi/\lambda) \times \Delta\ (\text{path difference})$$

Δ = path difference; λ = wavelength

Constructive: $\Delta = n\lambda \quad (n = 0, 1, 2, \ldots)$

Destructive: $\Delta = (2n - 1)\lambda/2$

3. Thin Film Interference (Reflection Geometry)

3.1 Phase Change on Reflection

Rule: When light reflects from a denser medium (higher refractive index), it undergoes a phase change of π (equivalent to an extra path of λ/2). No phase change when reflecting from rarer medium.

3.2 Optical Path Difference in a Thin Film

$$\Delta = 2\ \mu\ t\ \cos\ r$$

where μ = refractive index of film, t = thickness, r = angle of refraction inside film

3.3 Conditions (Air above & below film, normal incidence)

For Reflected Light:

Maxima (bright): $2\mu t = (2n - 1)\lambda/2$

Minima (dark): $2\mu t = n\lambda$

For Transmitted Light:

Maxima (bright): $2\mu t = n\lambda$

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Minima (dark): 2μt = (2n − 1)λ/2
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Note: Reflected and transmitted patterns are *complementary* – where reflected is bright, transmitted is dark, and vice versa.

3.4 Colours in Thin Films

- White light contains multiple wavelengths. For a given thickness t and angle, only certain wavelengths satisfy the maxima condition → those colours appear bright.
- As thickness varies, different colours are reinforced → rainbow-like colour patterns.
- Very thin films appear black in reflected light ($2\mu t \approx 0$ → destructive for all λ).

4. Newton's Rings

4.1 Setup & Principle

A plano-convex lens is placed on a flat glass plate. An air wedge of varying thickness t forms between them. Monochromatic light incident normally produces concentric circular fringes (Newton's rings) due to thin-film interference in the air gap.

4.2 Air Film Thickness at Radius r

$$t = r^2 / (2R)$$

R = radius of curvature of the plano-convex lens; r = radius at that point

4.3 Path Difference (Reflected, Air Film)

$$\Delta = 2t + \lambda/2 \quad (\lambda/2 \text{ due to phase change at lower glass surface})$$

4.4 Radius of n-th Ring

Bright ring: $r_n^2 = (2n - 1)\lambda R/2$

n = 1, 2, 3, ...

Dark ring: $r_n^2 = n\lambda R$

n = 0, 1, 2, ... (centre is dark)

4.5 Diameter of n-th Dark Ring

$$D_n^2 = 4n\lambda R$$

$$D_n = 2r_n = 2\sqrt{(n\lambda R)}$$

4.6 Determination of Wavelength (λ)

Measure diameters of n-th and m-th dark rings:

$$\lambda = (D_n^2 - D_m^2) / [4(n - m)R]$$

Eliminates errors due to imperfect contact at centre (where a dark spot may be spread).

4.7 Determination of Refractive Index (μ) of a Liquid

Fill the gap between lens and plate with liquid of refractive index μ. Rings shrink. Use:

$$\mu = (D_n^2 - D_m^2)_{air} / (D_n^2 - D_m^2)_{liquid}$$

5. Interference vs Diffraction – Comparison Table

Property	Interference	Diffraction
Origin	Superposition of waves from two or more coherent sources	Bending of waves around edges/obstacles; superposition of secondary wavelets from same wavefront
Number of sources	Two or more discrete sources	Single source (extended slit or aperture)
Fringe width	All bright/dark fringes have equal width	Central maximum is twice as wide; side maxima are narrower
Intensity	All bright fringes have equal intensity	Intensity decreases progressively from centre outward
Dark fringes	Perfectly dark (zero intensity)	Nearly dark (not perfectly zero in general)
Key formula	$\Delta = n\lambda$ (bright); $\Delta = (2n-1)\lambda/2$ (dark)	$a \sin\theta = n\lambda$ (dark minima, single slit)

6. Symbol Legend (Interference)

- λ – wavelength
- μ – refractive index
- t – film thickness
- r – angle of refraction
- R – radius of curvature
- n – order number
- D_n – diameter of n-th ring
- Δ – path difference
- δ – phase difference
- I – intensity
- r_n – radius of n-th ring
- m – ring number (reference)

7. Diffraction – Basics

Diffraction is the bending of light waves around obstacles or through apertures, caused by the interference of secondary wavelets (Huygens' Principle) emanating from different parts of the same wavefront.

7.1 Fresnel vs Fraunhofer Diffraction

Feature	Fresnel Diffraction	Fraunhofer Diffraction
Source distance	Finite (source near the slit)	Infinite (or effectively at infinity via lens)
Screen distance	Finite (screen near the slit)	Infinite (or at focal plane of converging lens)
Incident wavefront	Spherical or cylindrical	Plane wavefront
Lenses required	Not required	Required (to convert spherical to plane waves)
Mathematical complexity	Complex (involves Fresnel integrals)	Simpler (Fourier transform of aperture)
Practical example	Diffraction around a straight edge	Diffraction grating; single slit in lab

8. Fraunhofer Diffraction – Single Slit

8.1 Setup

Slit width = a; monochromatic light of wavelength λ incident normally. Pattern observed at focal plane of a converging lens.

8.2 Condition for Minima (Dark Fringes)

$$a \sin\theta = n\lambda \qquad n = \pm1, \pm2, \pm3, \ldots$$

a = slit width; θ = angle of diffraction; n = order of minimum ($n \neq 0$)

8.3 Intensity Distribution

$$I(\theta) = I_0 (\sin \alpha / \alpha)^2 \qquad \text{where } \alpha = \pi a \sin\theta / \lambda$$

- Central maximum at $\theta = 0$ ($I = I_0$), width = $2\lambda/a$.
- Secondary maxima (approx.) at $a \sin\theta = (2n+1)\lambda/2$.
- Intensities of 1st, 2nd, 3rd secondary maxima relative to central: 4.7%, 1.7%, 0.8%.

8.4 Angular Half-Width of Central Maximum

$$\sin\theta = \lambda/a \rightarrow \theta \approx \lambda/a \text{ (for small } \theta)$$

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9. Fraunhofer Diffraction – Double Slit

9.1 Combined Pattern

Double-slit pattern = Single-slit diffraction envelope × Two-slit interference fringes.

$$I(\theta) = I_0 (\cos \beta)^2 (\sin \alpha / \alpha)^2$$

where $\alpha = \pi a \sin\theta/\lambda$ (diffraction term) and $\beta = \pi d \sin\theta/\lambda$ (interference term)

a = slit width; d = slit separation (centre to centre)

9.2 Missing Orders

An interference maximum is absent (missing) when it coincides with a diffraction minimum:

$$d/a = n/m \rightarrow \text{order } n \text{ missing when } d = m \times a$$

e.g., if d = 2a, orders n = 2, 4, 6, ... are missing.

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10. Fraunhofer Diffraction – N-Slits (Diffraction Grating)

10.1 Principal Maxima Condition

$$(a + b)\sin\theta = n\lambda \quad n = 0, 1, 2, 3, \ldots$$

a = slit width; b = opaque portion; (a + b) = grating element d

10.2 Secondary Maxima and Minima

- Between consecutive principal maxima, there are (N − 1) minima and (N − 2) secondary maxima.
- As N increases, principal maxima become sharper and brighter.

10.3 Intensity Envelope

$$I = I_0\,(\sin N\gamma / N\sin\gamma)^2\,(\sin\alpha/\alpha)^2 \quad \text{where } \gamma = \pi(a+b)\sin\theta/\lambda$$

11. Diffraction Grating – Key Formulas

11.1 Grating Equation

$$(a + b)\sin\theta = n\lambda \quad \text{OR} \quad d\sin\theta = n\lambda$$

d = grating element = (a + b); n = order of principal maximum; θ = diffraction angle

11.2 Grating Element

$d = 1 / N$ (N = number of lines per unit length, in m^{-1} or cm^{-1})

11.3 Grating Summary Table

Term	Symbol	Formula / Meaning
Grating element	$d = (a+b)$	Distance between corresponding points of adjacent slits
Order of spectrum	n	n = 0 (central white), n = 1, 2, ... (coloured spectra)
Max order possible	n_{max}	$n_{max} = d/\lambda$ ($\sin\theta \leq 1$)
Grating equation	$d\sin\theta = n\lambda$	Gives angle of n-th order principal maximum

11.4 Dispersive Power (Qualitative)

Definition: Dispersive power = dθ/dλ – the angular separation produced per unit wavelength difference.

$$d\theta/d\lambda = n / (d\cos\theta)$$

- Increases with order n (higher orders disperse more).
- Increases as grating element d decreases (finer grating).
- Increases as θ increases (cosθ smaller).

11.5 Resolving Power (Qualitative)

Definition: Resolving power = λ/dλ – ability to distinguish two closely spaced spectral lines (λ and λ + dλ).

$$R = \lambda/d\lambda = nN$$

n = order; N = total number of slits (grating lines) illuminated

- Increases with total number of slits N (wider grating or more slits).
- Increases with order n.
- Does NOT depend on grating element d directly (only via N).

Dispersive Power ↑ when:

- Order n is higher
- Grating element d is smaller
- Diffraction angle θ is larger

Resolving Power ↑ when:

- Number of slits N is larger
- Order n is higher
- Grating width is larger

12. Symbol Legend (Diffraction & Grating)

- a – slit width
- b – opaque width
- d – grating element
- N – total slits
- θ – diffraction angle
- n – order
- λ – wavelength
- dλ – wavelength diff.

- α – πa sinθ/λ
- β – πd sinθ/λ
- I_0 – central intensity
- R – resolving power

13. Polarization – Basics

Light is a transverse electromagnetic wave. In ordinary (unpolarized) light, the electric field vector vibrates in all planes perpendicular to propagation. **Polarization** is the process by which vibrations are restricted to a single plane (or a specific pattern).

13.1 Types of Polarized Light

Type	Electric Field Behaviour	Condition	Example / Production
Linear (Plane)	Oscillates in a single fixed plane	Two components in phase ($\delta = 0$ or $n\pi$)	Reflection, refraction, Nicol's prism
Circular	Tip traces a circle; magnitude constant	Equal amplitudes; $\delta = \pm\pi/2$	Quarter wave plate on linearly polarized light
Elliptical	Tip traces an ellipse	Unequal amplitudes OR $\delta \neq \pi/2$	General case; QWP with unequal components
Unpolarized	Vibrates in all directions randomly	No fixed phase relationship	Ordinary bulb, sunlight

13.2 Methods of Polarization

By Reflection

At Brewster's angle i_B, reflected ray is completely plane polarized.

$$\tan i_B = \mu$$

(Brewster's Law)

Reflected + refracted rays are perpendicular: $i_B + r = 90°$

By Refraction

Light transmitted through a pile of glass plates becomes partially polarized (refracted ray has excess of vibrations in the plane of incidence).

More plates → more polarization.

By Double Refraction (Birefringence)

Anisotropic crystals (calcite, quartz) split light into two rays: ordinary (O-ray) and extraordinary (E-ray), both plane polarized at 90° to each other.

13.3 Malus's Law

$$I = I_0 \cos^2\theta$$

I_0 = intensity of incident polarized light; θ = angle between polarizer and analyser axes; I = transmitted intensity

- $\theta = 0°$ → $I = I_0$ (maximum transmission)
- $\theta = 90°$ → $I = 0$ (complete extinction)

14. Nicol's Prism

14.1 Construction (Conceptual)

- Made from a calcite crystal (Iceland spar), length ~3× its width.
- Crystal is cut along a diagonal and the two halves are cemented with Canada balsam (refractive index 1.55, between $\mu_O = 1.658$ and $\mu_E = 1.486$).
- End faces are cut so that the optic axis makes specific angles.

14.2 Working Principle

- Unpolarized light enters and splits into O-ray ($\mu_O = 1.658$) and E-ray (μ_E varies).
- O-ray strikes the Canada balsam layer at an angle exceeding its critical angle → undergoes Total Internal Reflection (TIR) → removed from the beam.
- E-ray passes through the Canada balsam → emerges as plane polarized light.
- Critical angle for O-ray in Canada balsam = $\sin^{-1}(1.55/1.658) \approx 69°$.

14.3 Uses

- As a polarizer: produces plane polarized light from unpolarized light.

- As an analyser: detects whether light is polarized and determines the plane of polarization.
- Pair of Nicols: used in polarimeters, saccharimeters, petrographic microscopes.

Crossed Nicols: When polarizer and analyser are oriented at 90° to each other, no light passes through → complete darkness (extinction). Rotating one Nicol by 90° restores full transmission.

15. Wave Plates (Retardation Plates)

Wave plates are cut from uniaxial birefringent crystals (e.g., calcite, quartz) such that the optic axis lies in the plane of the plate. They introduce a controlled phase difference (δ) between the O-ray and E-ray passing through them.

15.1 Path Difference and Phase Difference

$$\text{Path difference} = (\mu_E - \mu_O) \times t$$

$$\text{Phase difference } \delta = (2\pi/\lambda) \times (\mu_E - \mu_O) \times t$$

t = thickness of wave plate; μ_O, μ_E = refractive indices for O-ray and E-ray

15.2 Half Wave Plate (HWP) vs Quarter Wave Plate (QWP)

Feature	Half Wave Plate (HWP)	Quarter Wave Plate (QWP)
Phase difference introduced	$\delta = \pi$ (180°)	$\delta = \pi/2$ (90°)
Path difference	$\lambda/2$	$\lambda/4$
Thickness	$t = \lambda / [2(\mu_O - \mu_E)]$	$t = \lambda / [4(\mu_O - \mu_E)]$
Effect on linearly polarized light	Rotates the plane of polarization by 2θ (where θ = angle with optic axis)	Produces elliptically polarized light in general; circularly polarized when θ = 45°
Effect on circularly polarized light	Converts circular to linear polarization	Converts circular to linear polarization
Special case	If θ = 45°, plane of polarization rotates by 90° (perpendicular to original)	If incident light has equal components at 45°, output is circularly polarized
Key use	Rotating polarization direction; converting polarization states	Producing/analysing circular and elliptical polarization

15.3 Thickness Formulas

$$\text{HWP thickness: } t_{HWP} = \lambda / [2|\mu_O - \mu_E|]$$

$$\text{QWP thickness: } t_{QWP} = \lambda / [4|\mu_O - \mu_E|]$$

16. Detection of Polarization States

Test with Rotating Analyser	Observation	Light Type
Intensity varies; zero at 2 positions in 360°	Complete extinction twice per rotation	Plane (linearly) polarized
Intensity constant; no variation	Same intensity at all angles	Circularly polarized OR unpolarized
Intensity varies; never zero	Max and min but not zero	Elliptically polarized OR partially polarized

To distinguish circular from unpolarized: Insert a QWP before the analyser. If light becomes linearly polarized (extinction occurs on rotating analyser), original was circularly polarized. If still no extinction, original was unpolarized.

17. Symbol Legend (Polarization)

- μ_O – refractive index (O-ray)
- μ_E – refractive index (E-ray)
- i_B – Brewster's angle
- δ – phase difference
- t – plate thickness

θ – angle with optic axis

I_0 – incident intensity

I – transmitted intensity

18. Flashcards – Quick Recall

Q: What is interference?

A: Redistribution of light intensity due to superposition of two or more coherent waves. Energy is neither created nor destroyed – only redistributed.

Q: Condition for constructive interference?

A: Path difference $\Delta = n\lambda$ ($n = 0, 1, 2, \ldots$) or phase difference $\delta = 2n\pi$

Q: Condition for destructive interference?

A: Path difference $\Delta = (2n - 1)\lambda/2$ or phase difference $\delta = (2n - 1)\pi$

Q: Why is the centre of Newton's rings dark in reflected light?

A: At the point of contact, $t = 0$, so $\Delta = \lambda/2$ (due to phase change at lower surface only) → destructive interference → dark spot.

Q: Diameter of n-th dark ring in Newton's rings?

A: $D_n^2 = 4n\lambda R \Rightarrow D_n = 2\sqrt{(n\lambda R)}$

Q: Formula for wavelength from Newton's rings?

A: $\lambda = (D_n^2 - D_m^2) / [4(n - m)R]$

Q: What is diffraction?

A: Bending of light waves around the edges of an obstacle or aperture, explained by Huygens' principle and secondary wavelets from the same wavefront.

Q: Diffraction minimum condition (single slit)?

A: $a \sin\theta = n\lambda$ ($n = \pm1, \pm2, \ldots$) where a = slit width, θ = angle of minimum

Q: How does slit width affect diffraction?

A: Smaller slit → wider central diffraction maximum. As a ↓, angular width $2\lambda/a$ ↑. For $a \to \lambda$, light spreads in all directions.

Q: Grating equation?

A: $d \sin\theta = n\lambda$ where $d = (a+b)$ = grating element, n = order

Q: Resolving power of a grating?

A: $R = \lambda/d\lambda = nN$ where n = order, N = total number of slits illuminated

Q: Dispersive power of a grating?

A: $d\theta/d\lambda = n/(d \cos\theta)$; increases with order n and decreases with grating element d

Q: What is polarization of light?

A: The process of restricting the vibrations of the electric field vector of light to a single plane (or specific pattern), confirming the transverse wave nature of light.

Q: State Brewster's Law.

A: $\tan i_B = \mu$; at Brewster's angle i_B, reflected ray is completely plane polarized, and $i_B + r = 90°$

Q: State Malus's Law.

A: $I = I_0 \cos^2\theta$; intensity of transmitted light through an analyser depends on $\cos^2$ of angle between polarizer and analyser axes.

Q: Purpose of a quarter wave plate?

A: Introduces a phase difference of $\pi/2$ (path diff. = $\lambda/4$). Converts linearly polarized light to circularly polarized (when $\theta = 45°$) or elliptically polarized light.

Q: Purpose of a half wave plate?

A: Introduces a phase difference of π (path diff. = $\lambda/2$). Rotates the plane of polarization by 2θ. If $\theta = 45°$, the plane rotates by $90°$.

Q: How does Nicol's prism work?

A: O-ray undergoes TIR at the Canada balsam layer (exceeds critical angle) and is eliminated. E-ray passes through → emerges as plane polarized light.

Q: Difference between O-ray and E-ray?

A: O-ray obeys Snell's law and has constant μ_O in all directions. E-ray does not obey Snell's law; μ_E varies with direction of propagation.

Q: Condition for producing circular polarization with a QWP?

A: Incident linearly polarized light must have equal amplitudes of its components along the fast and slow axes of the QWP (i.e., angle $\theta = 45°$ with optic axis).

Q: What is a missing order in double-slit diffraction?

A: An interference maximum absent because it coincides with a diffraction minimum. Occurs when d/a = integer ratio. e.g., $d = 2a$ → 2nd, 4th, ... orders missing.

Q: Why does Newton's rings pattern have circular fringes?

A: The air gap thickness t is constant along circles ($t = r^2/2R$) due to the spherical lens surface. Equal thickness → equal path difference → circular fringes.

Q: Relationship between Δ (path diff.) and δ (phase diff.)?

A: $\delta = (2\pi/\lambda) \times \Delta$

Q: Thin film – condition for maxima in reflected light?

A: $2\mu t \cos r = (2n - 1)\lambda/2$ (includes $\lambda/2$ shift due to reflection from denser medium)

Q: What is the intensity formula for N-slit diffraction?

A: $I = I_0 (\sin N\gamma/N \sin \gamma)^2 (\sin \alpha/\alpha)^2$ where $\gamma = \pi d \sin\theta/\lambda$

19. Exam Tips & Common Mistakes to Avoid

19.1 Interference – Thin Films

- **Phase change on reflection:** Always check which surface causes the λ/2 shift. Phase change occurs ONLY when reflecting from a denser medium. If both surfaces give a phase change, they cancel out. If only one gives a phase change, it adds λ/2 to path difference. Missing this is the most common error in thin film problems.
- **Path difference vs optical path difference:** The geometric path difference is 2t cos r; the optical path difference is 2μt cos r. Always use the optical path difference (μ included) for interference conditions.
- **Reflected vs transmitted conditions:** They are complementary. Maxima in reflected = minima in transmitted and vice versa. Do not use the same formula for both without checking.

19.2 Newton's Rings

- **Centre is always dark in reflected light:** At t = 0, path difference = λ/2 (phase change) → destructive interference. Students sometimes write that the centre is bright; that is for transmitted light.
- **Use diameter difference formula for λ:** Never use a single ring's diameter to find λ, because the contact point may not be perfect (slightly raised or depressed). Always use $D_n^2 - D_m^2$ to eliminate the error.

19.3 Diffraction

- **Interference vs Diffraction minimum conditions:** For double-slit interference, the condition for dark fringes uses d (slit separation). For single-slit diffraction minima, the condition uses a (slit width). Do not confuse these: a sinθ = nλ is diffraction; d sinθ = (2n−1)λ/2 is interference dark fringe.
- **Central maximum width:** The first diffraction minimum is at sinθ = λ/a. The angular half-width is λ/a, so the full central maximum spans 2λ/a. It is twice as wide as secondary maxima.
- **Grating – maximum order:** $n_{max} = d/\lambda$ (since sinθ ≤ 1). Do not attempt to calculate orders higher than this; they physically cannot exist.

19.4 Grating Problems

- **Convert lines/cm or lines/mm to grating element d:** If 500 lines/mm, then d = 1/500 mm = 2×10^{-3} mm = 2×10^{-6} m. Always work in consistent SI units (metres and metres).
- **Resolving power depends on N (total slits), not d:** Two gratings with same d but different widths have different resolving powers. R = nN, not n/d.

19.5 Polarization

- **Circular vs elliptical polarization:** Circular requires equal amplitudes AND phase difference = π/2. If amplitudes are unequal, even with δ = π/2, the result is elliptical. If amplitudes are equal but δ ≠ π/2, also elliptical.
- **Malus's Law applies only to incident polarized light:** If the incident light is unpolarized, rotating the analyser gives $I = I_0/2$ (constant, not varying as $\cos^2\theta$). Malus's law is used only when the incident beam is already polarized.
- **Crossed Nicols:** When two Nicols are crossed (90°), I = 0. If a birefringent plate is placed between them at an angle, light may pass through due to the new polarization components introduced. This is a common conceptual question.

General Strategy for Numerical Problems: Write down the given data, identify the formula clearly, convert all units to SI (metres, radians), substitute, and verify the order of magnitude of the answer. For optics problems, wavelengths are typically 400–700 nm = 4×10^{-7} to 7×10^{-7} m.

20. Quick Revision Sheet – All Formulas (No Explanations)

SUPERPOSITION & INTERFERENCE

$I = I_1+I_2+2\sqrt{(I_1 I_2)}\ \cos\delta$

$I_{max} = (\sqrt{I_1}+\sqrt{I_2})^2$

$I_{min} = (\sqrt{I_1}-\sqrt{I_2})^2$

$\delta = (2\pi/\lambda)\Delta$

Bright: $\Delta = n\lambda$

Dark: $\Delta = (2n-1)\lambda/2$

THIN FILM (REFLECTED)

Path diff.: $2\mu t \cos r$

Bright: $2\mu t \cos r = (2n-1)\lambda/2$

Dark: $2\mu t \cos r = n\lambda$

NEWTON'S RINGS

$t = r^2/2R$

Dark ring: $D_n^2 = 4n\lambda R$

Bright ring: $D_n^2 = 2(2n-1)\lambda R$

$\lambda = (D_n^2-D_m^2)\ /\ 4(n-m)R$

$\mu = (D_n^2-D_m^2)_{air}\ /\ (D_n^2-D_m^2)_{liq}$

DIFFRACTION – SINGLE SLIT

Minima: $a \sin\theta = n\lambda$

$I(\theta) = I_0(\sin\alpha/\alpha)^2$

$\alpha = \pi a \sin\theta/\lambda$

Central max half-width: $\sin\theta = \lambda/a$

DIFFRACTION – DOUBLE SLIT

$I = I_0(\cos\beta)^2(\sin\alpha/\alpha)^2$

$\beta = \pi d \sin\theta/\lambda$

Missing order: d/a = integer

DIFFRACTION GRATING

$d \sin\theta = n\lambda$

$d = 1/N$ (N = lines/m)

$n_{max} = d/\lambda$

Dispersive power: $d\theta/d\lambda = n/(d \cos\theta)$

Resolving power: $R = nN$

POLARIZATION

Malus: $I = I_0\cos^2\theta$

Brewster: $\tan i_B = \mu$

$i_B + r = 90°$

WAVE PLATES

Path diff. $= (\mu_E-\mu_O)t$

Phase diff. $= (2\pi/\lambda)(\mu_E-\mu_O)t$

HWP: $t = \lambda/[2|\mu_O-\mu_E|]$; $\delta = \pi$

QWP: $t = \lambda/[4|\mu_O-\mu_E|]$; $\delta = \pi/2$

N-SLIT INTENSITY

$I = I_0(\sin N\gamma/N\sin\gamma)^2(\sin\alpha/\alpha)^2$

$\gamma = \pi(a+b)\sin\theta/\lambda$

21. Master Comparison Tables

21.1 All Interference Conditions at a Glance

Situation	Condition for Maxima (Bright)	Condition for Minima (Dark)
General (any geometry)	$\Delta = n\lambda$	$\Delta = (2n-1)\lambda/2$
Thin film (reflected, 1 phase change)	$2\mu t \cos r = (2n-1)\lambda/2$	$2\mu t \cos r = n\lambda$
Thin film (transmitted)	$2\mu t \cos r = n\lambda$	$2\mu t \cos r = (2n-1)\lambda/2$
Newton's rings (reflected, dark)	$r_n^2 = (2n-1)\lambda R/2$	$r_n^2 = n\lambda R$ $(D_n^2 = 4n\lambda R)$

21.2 Diffraction Patterns Summary

Type	Minima Condition	Maxima (approx.)	Central Max Width
Single slit	$a \sin\theta = n\lambda$ $(n \neq 0)$	$a \sin\theta = (2n+1)\lambda/2$	$2\lambda/a$
Double slit (interference)	$d \sin\theta = (2n-1)\lambda/2$	$d \sin\theta = n\lambda$	λ/d (fringe width)
Grating (N slits)	Principal maxima: $d \sin\theta = n\lambda$	Sharper & brighter for large N	Grating equation gives θ

21.3 Wave Plates Comparison (Expanded)

Property	HWP (Half Wave Plate)	QWP (Quarter Wave Plate)
Phase retardation	π	$\pi/2$
Path difference	$\lambda/2$	$\lambda/4$
Thickness	$\lambda / 2\lvert\mu_O - \mu_E\rvert$	$\lambda / 4\lvert\mu_O - \mu_E\rvert$
Linear → ?	Linear (rotated by 2θ)	Elliptical ($\theta \neq 45°$) or Circular ($\theta = 45°$)
Circular → ?	Circular (opposite handedness)	Linear
Elliptical → ?	Elliptical (axes rotated)	Elliptical (modified)

21.4 Polarization Types – Full Comparison

Type	Amplitude Condition	Phase Condition	Locus of E-vector tip
Linear	Any ratio	$\delta = 0$ or $n\pi$	Straight line
Circular (right)	$E_x = E_y$	$\delta = +\pi/2$	Circle (anticlockwise when facing source)
Circular (left)	$E_x = E_y$	$\delta = -\pi/2$	Circle (clockwise when facing source)
Elliptical	$E_x \neq E_y$ or any $\delta \neq 0, \pi, \pi/2$	General δ	Ellipse
Unpolarized	All directions	No fixed δ	Random in all directions

21.5 Fresnel vs Fraunhofer – Quick Comparison

Aspect	Fresnel	Fraunhofer
Wavefront	Spherical / cylindrical	Plane
Distances	Finite (source & screen)	Infinite (lenses used)
Maths	Fresnel integrals (complex)	Fourier / simpler
Practical use	Edge diffraction, zone plates	Gratings, slits in labs

22. Formula Master Reference (All in One Place)

Superposition

$y = y_1 + y_2$

$I = I_1+I_2+2\sqrt{I_1 I_2}\cos\delta$

$I_{max} = (\sqrt{I_1}+\sqrt{I_2})^2$

$I_{min} = (\sqrt{I_1}-\sqrt{I_2})^2$

Path/Phase

$\delta = (2\pi/\lambda)\Delta$

Bright: $\Delta = n\lambda$

Dark: $\Delta=(2n-1)\lambda/2$

Thin Film

$2\mu t \cos r$

Bright: $(2n-1)\lambda/2$

Dark: $n\lambda$

(reflected, 1 phase change)

Newton's Rings

$t = r^2/2R$

Dark: $D_n^2=4n\lambda R$

$\lambda=(D_n^2-D_m^2)/4(n-m)R$

$\mu=D_{air}^2/D_{liq}^2$

Single Slit

$a \sin\theta=n\lambda$ (dark)

$I=I_0(\sin\alpha/\alpha)^2$

$\alpha=\pi a \sin\theta/\lambda$

Double Slit

$I=I_0(\cos\beta)^2(\sin\alpha/\alpha)^2$

$\beta=\pi d \sin\theta/\lambda$

Missing: d/a=integer

Grating

$d \sin\theta=n\lambda$

$d=1/N$

$n_{max}=d/\lambda$

$R=nN$

$d\theta/d\lambda=n/(d \cos\theta)$

Polarization

$I=I_0\cos^2\theta$

$\tan i_B=\mu$

$i_B+r=90°$

Wave Plates

HWP: $\delta=\pi$; $t=\lambda/2(\mu_O-\mu_E)$

QWP: $\delta=\pi/2$; $t=\lambda/4(\mu_O-\mu_E)$

23. Key Physical Insights – Must Know

Why Newton's rings are circular?

Locus of equal thickness in the air gap is a circle (symmetric about point of contact). Equal thickness → equal path difference → circular fringes.

Why grating produces sharp maxima?

As N ↑, constructive interference from N slits reinforces at specific angles, while destructive interference occurs everywhere else → sharper, brighter lines.

Why is light transverse?

Only transverse waves can be polarized. Since light can be polarized, it is a transverse wave. Longitudinal waves (sound) cannot be polarized.

Coherence requirement?

Interference is observable only when path difference ≤ coherence length of the source. Laser has very large coherence length; ordinary sources are limited.

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24. Important Constants & Typical Values

Visible λ: 400–700 nm
Yellow (Na): 589.3 nm
Speed of light c: 3×10^{8} m/s
1 nm = 10^{-9} m
1 Angstrom = 10^{-10} m
μ glass ≈ 1.5
μ calcite O: 1.658
μ calcite E: 1.486
μ Canada balsam: 1.55
μ water: 1.33
Brewster θ for glass: ~56°
Critical angle (calcite/balsam): ~69°

Last-minute reminder: Wave Optics is built on three pillars – *Interference* (path difference & coherence), *Diffraction* (bending around edges), and *Polarization* (transverse nature). Every formula in this sheet connects back to one of these three ideas. Understand the physics, not just the formula.
